

Segment based on pointing a few points as positives and negatives:

https://github.com/facebookresearch/sam2/blob/main/notebooks/image_predictor_example.ipynb

other stuff (textual inputs):

<https://github.com/IDEA-Research/GroundingDINO>

<https://github.com/IDEA-Research/Grounded-SAM-2>

<https://github.com/LLaVA-VL/LLaVA-NeXT>

We train a neural network for classification.

We get an accuracy of $0.93 \sim 93\%$ on test data and of 0.94 on training data.

What can we do with this information ?

You are in your PhD. You train another neural network for classification.

You get an accuracy of $0.93 \sim 93\%$ on test data and this is higher than the results with other methods. Yay, a paper will be written :) .

What can you do with this information ?

next: what is a good machine learning/deep learning model ?

The goal of machine learning described in words

To find a prediction model, which predicts well(=?) on unseen and unforeseeable data points

two things to be clarified:

- formalize: on unseen and unforeseeable data points
... using probability distributions.
- formalize: well
... using mean and expectation

- 1 Probability distributions
- 2 Modelling future unforeseeable data points
- 3 The goal of training machine learning / AI systems
- 4 Why one MUST NEVER use test data to select/optimize your ML system
- 5 Overfitting

the first idea:

aka Mittelwert über eine Stichprobe

Mean over data samples / Mittelwert über eine Stichprobe

given n samples $x^{\{i\}}$ from a vector space: $S = (x^{\{0\}}, \dots, x^{\{n-1\}})$,

$$M_1(S) := \frac{1}{n} \sum_{i=0}^{n-1} x^{\{i\}}$$

Beispiel:

$$\{(3, 0, 1), (-2, 4, -5), (6, -1, 3)\}, n = 3$$

$$M_1(S) = (7/3, 1.0, -1/3)$$

random variable X :

- concept: a mathematical model for a process which outputs values $X(o)$ in a non-predictable manner
- we will characterize the process by probabilities for its outputs

no exam stuff/kein Klausurstoff:

- technically: a function $X : O \rightarrow E$ which maps outcomes $o \in O$ (O - Ergebnisraum, $o \in O$ zufälliges Ereignis) of a randomized experiment onto numbers in E
- we omit here further details on the concept of measurability. This requires to define sets of measurable events for O and E

discrete probability distribution

- we have a set $S = \{x^{\{0\}}, \dots, x^{\{i\}}, \dots\}$ which is finite (endlich viele) or countably infinite (abzählbar unendlich)
- a discrete probability is a mapping $P : S \rightarrow [0, 1]$ with the following properties:

$$x^{\{i\}} \in S \Rightarrow P(x^{\{i\}}) \in [0, 1]$$

$$\sum_{x^{\{i\}} \in S} P(x^{\{i\}}) = 1$$

Examples:

$$S = \{0, \dots, 5\}, P(i) = \frac{1}{6}$$

$$S = \{0, \dots, 5\}, P(i) = \frac{i+1}{21}$$

$$S = \{i \in \mathbb{Z}, i \geq 1\}, P(i) = 2^{-i}$$

$$S = \{i \in \mathbb{Z}, i \geq 1\}, P(i) = \frac{6}{i^2 \pi^2}$$

- whole number/ganze Zahlen are countable
- $\mathbb{Q} = \{\frac{p}{q}, p \in \mathbb{Z}, q \in \mathbb{Z}, q \geq 1\}$ (rationale Zahlen) is countable
- $\mathbb{R}$ (reelle Zahlen) is **not countable** (no exam / kein Klausurstoff:
https://www.mathe-online.at/mathint/zahlen/i_Rueberabz.html)

absolutely continuous probability distribution

- we have an interval $I \subset \mathbb{R}$
- we want to model the probability that a random variable X takes values in an interval $[a, b] \subset I$
- an absolutely continuous probability distribution $P : I \rightarrow [0, 1]$ is defined by a density function $f : I \rightarrow \mathbb{R}$ such that:

$$f(x) \geq 0$$

$$[a, b] \subset I \Rightarrow P(X \in [a, b]) = \int_{x=a}^{x=b} f(x) dx$$

$$P(X \in I) = 1 = \int_{x \in I} f(x) dx$$

Conceptually, integration and measure theory is the generalization of summing to non-countable sets like intervals.

Examples:

$$I = [0, 1], f(x) = 1 \Rightarrow P(X \in [a, b]) = b - a$$

$$I = [0, 1], f(x) = 3x^2 \Rightarrow P(X \in [a, b]) = b^3 - a^3$$

$$I = [0, 2\pi], f(x) = 1[x \leq \pi] \frac{1}{2} \sin(x)$$

$$\text{if } a < \pi \Rightarrow P(X \in [a, b]) = -\frac{1}{2} \cos(\min(b, \pi)) + \frac{1}{2} \cos(a)$$

$$P(X \in [0, 2\pi]) = -\frac{1}{2} \cos(\pi) + \frac{1}{2} \cos(0) = -(-1/2) + 1/2$$

$$I = \mathbb{R}, f(x) = \frac{1}{\pi(1+x^2)}$$

aka Erwartungswert über eine Verteilung

Expectation over a distribution / Erwartungswert über eine Verteilung, discrete case

given a set of values $S = \{x^{\{0\}}, \dots, x^{\{i\}}, \dots\}$ from a vector space which possibly could be outputted by a random variable, we define

$$E[X] := \sum_i x^{\{i\}} P(x^{\{i\}})$$

Beispiel:

aka Erwartungswert über eine Verteilung

Expectation over a distribution / Erwartungswert über eine Verteilung, case with a density function

given an interval I ,

$$E[X] := \int_{x \in I} x f(x) dx$$

Beispiel:

$$I = [0, 1], f(x) = 1, E[x] = \int_0^1 x dx = \frac{x^2}{2} \Big|_{x=0}^{x=1} = \frac{1^2}{2} - \frac{0^2}{2}$$

note the analogy:

$$E[X] := \sum_i x^{\{i\}} P(x^{\{i\}})$$

$$\sum_i (\dots) \leftrightarrow$$

$$P(x^{\{i\}}) \leftrightarrow$$

$$E[X] := \int_{x \in I} x f(x) dx$$

$$\int_{ix \in I} (\dots) dx$$

$$f(x)$$

Conceptually, integration and measure theory is the generalization of summing to non-countable sets like intervals.

How are they related ?

- mean is defined over the samples which we have collected
- expectation is defined over all possible values which can be observed as a sample

How are they related ?

- mean is defined over the samples which we have collected
- the mean over a collection of samples S can be defined as an expectation over a space T which contains (**only**) all unique samples in S !
(and the probability is zero for all values not observed in S)

$$S = (x^{\{0\}}, \dots, x^{\{n-1\}})$$

$$T = \{x^{\{i\}} \in S\} \text{ here every sample from } S \text{ appears only 1x}$$

$$\text{count}(S)_T[x] = \text{the number of occurrences of } x \text{ in } S$$

$$= \sum_{i=0}^{n-1} \mathbb{1}[x = x^{\{i\}}]$$

$$t \in T \Rightarrow P(X = t) = \frac{\text{count}(S)_T[x]}{n} = \frac{\text{how often does } t \text{ occur in } S}{\text{number of elements in } S}$$

How are they related ?

Then we have:

$$\begin{aligned}
 1 &= \sum_{t \in T} 1[x^{\{i\}} == t] \\
 \frac{1}{n} \sum_{i=0}^{n-1} x^{\{i\}} &= \frac{1}{n} \sum_{i=0}^{n-1} \sum_{t \in T} x^{\{i\}} 1[x^{\{i\}} == t] \\
 &= \frac{1}{n} \sum_{i=0}^{n-1} \sum_{t \in T} x^{\{i\}} 1[x^{\{i\}} == t] \\
 &= \frac{1}{n} \sum_{i=0}^{n-1} \sum_{t \in T} t 1[x^{\{i\}} == t] \text{ switch now the summings!} \\
 &= \sum_{t \in T} t \frac{1}{n} \sum_{i=0}^{n-1} 1[x^{\{i\}} == t] = \sum_{t \in T} t \frac{\text{count}(S)_T[x]}{n} \\
 &= \sum_{t \in T} t P(X = t)
 \end{aligned}$$

Summary: the mean over n samples in S is an expectation over $T = \{ \text{unique values} \}$ in S with a probability $P(X = t), t \in T$ such that

⊙ ...

$$\begin{aligned}\Rightarrow P(X = t) &= \frac{\text{count}(S)_T[x]}{n} = \frac{\text{how often does } t \text{ occur in } S}{\text{number of elements in } S} \\ &= \frac{\sum_{i=0}^{n-1} \mathbb{1}[x^{\{i\}} == t]}{n}\end{aligned}$$

Note again: if $\hat{t} \notin S$ then $P(X = \hat{t}) = \frac{0}{n}$

The mean over a collection of samples S can be defined as an expectation over a space T which contains (only) all unique samples in S !

- ① Probability distributions
- ② Modelling future unforeseeable data points
- ③ The goal of training machine learning / AI systems
- ④ Why one MUST NEVER use test data to select/optimize your ML system
- ⑤ Overfitting

back to:

The goal of machine learning described in words

To find a prediction model, which **predicts well(=?)** on **unseen and unforeseeable data** points

steps:

- ⦿ model **on unseen and unforeseeable data** points by drawing them from a random variable which has a certain probability distribution
- ⦿ model **predicts well** by low loss value in expectation over the probability distribution of that random variable

We had before:

the goal of machine learning described in words

To find a prediction model, which predicts well on unseen and unforeseeable data points

What does it mean **on unseen and unforeseeable data points**?

- We do not know what samples (x, y) we can expect in the future
- uncertainty can be modeled by assuming that the samples are drawn from a probability distribution:

$$(x, y) \sim P(x, y)$$

A common assumption in machine learning described in words

Model the uncertainty of future/unseen data by assuming that there exists a probability distribution P which generates the data:

$$(x, y) \sim P(x, y)$$

Next step: conditional probabilities.

The use case for conditional probabilities are random processes, in which a pair (x, y) is drawn, and the value for X can be observed first, before we can observe the value for Y .

Simple example: $Y = 1$ – will get diabetes in the next 5 years, X = current age.

Y = Giftgehalt von Giftzwerge, X = location of the forest

intuition behind conditional probabilities

We consider a pair of random variables (X, Y) (which could have some kind of relationship).

$P(Y = y|X = x)$ is the probability that the variable Y takes value y given that we know that the other variable has a specific value $X = x$.

This is a probability for the values of one variable, given that we obtained information about the value of the other variable.

two variables:

$$P(Y = y|X = x) = \frac{P(Y = y, X = x)}{P(X = x)}$$

n variables:

$$P(X_0, X_1, \dots, X_n | X_{n+1}, \dots, X_{n+k}) = \frac{P(X_0, X_1, \dots, X_n, X_{n+1}, \dots, X_{n+k})}{P(X_{n+1}, \dots, X_{n+k})}$$

Let A, B be **sets of random variables**. Then we can write it compactly:

$$P(A = a | B = b) = \frac{P(A = a, B = b)}{P(B = b)}$$

How to memorize the definition?

how to remember the definition of conditional probability

- ⊙ requirement 1: $P(A = a|B = b) = C P(A = a, B = b)$, it is the joint probability times some constant
- ⊙ requirement 2: $P(A|B = b)$ is a Probability over the (non-conditioning) variables in A :

$$\sum_a P(A = a|B = b) = 1$$

From there one can easily deduce c and the definition of the conditional probability:

$$\begin{aligned} P(A = a|B = b) &= C P(A = a, B = b) \\ 1 &= \sum_a P(A = a|B = b) = C \sum_a P(A = a, B = b) = C P(B = b) \end{aligned}$$

solving for C : $C = \frac{1}{P(B=b)}$

next step: a brief look at statistical independence

The standard definition of statistical independence for discrete variables

Discrete variables $X_0, \dots, X_n$ are statistically independent if for all possible values $x_0, \dots, x_{n-1}$ the following holds:

$$P(X_0 = x_0, X_1 = x_1, \dots, X_{n-1} = x_{n-1}) = P(X_0 = x_0)P(X_1 = x_1) \cdots P(X_{n-1} = x_{n-1})$$

For the case with a density function a similar definition holds using the density functions.

The more interpretable of statistical independence: We have for all possible values $x_0, \dots, x_{n-1}$ that:

$$P(X_0, X_1, \dots, X_n | X_{n+1}, \dots, X_{n+k}) = P(X_0, X_1, \dots, X_n)$$

for two variables or two sets of variables:

$$\forall a, b : P(A = a | B = b) = P(A = a)$$

the meaning of statistical independence

A and B are independent random variables:

Knowing the exact value of B does not tell anything about the probability distribution of A

how to remember the definition of conditional probability

- requirement 1: $P(A = a|B = b) = cP(A = a, B = b)$
- requirement 2: $P(A|B = b)$ is a Probability over the variables in A , that is

$$\sum_a P(A = a|B = b) = 1$$

There is a second way to obtain a probability over only the variables in A given the joint probability $P(A = a, B = b)$, by computing the marginal probability / Randverteilung :

$$P(A = a) = \sum_{\text{all possible } b} P(A = a, B = b)$$

- question: How are

the conditional probability $P(A = a|B = b)$ and the marginal probability $P(A = a)$ related to each other?

$$\begin{aligned} P(A = a) &= \sum_{\text{all possible } b} P(A = a, B = b) = \sum_{\text{all possible } b} \frac{P(A = a, B = b)}{P(B = b)} P(B = b) \\ &= \sum_{\text{all possible } b} P(A = a|B = b)P(B = b) = E_{b \sim P(B)}[P(A = a|B = b)] \end{aligned}$$

So:

marginal vs conditional distribution

$$P(A = a|B = b) \overset{\text{compare}}{\longleftrightarrow} \overset{\text{vgl.}}{P(A = a)} = E_{b \sim P(B)}[P(A = a|B = b)]$$

The marginal distribution is an expectation over conditional distributions for all possible conditioning values !

Makes perfect sense: for the marginal distribution we don't know the value of the conditioning variables, so we average over all possible choices. For the conditional distribution we know the values, so we pick that values and compute the probability given that we know them.

Next step: an excursion to

- how can a random draw of samples $(x, y) \sim P(x, y)$ look like in practice?
- one common principle

$$p(x, y) = p(x|y)p(y)$$

$$p(x, y) = p(y|x)p(x)$$

Examples for such abstract $(x, y) \sim P(x, y)$ in binary classification:

- Input space $X = \mathbb{R}^1$, Output Space $\{-1, +1\}$

$$x \sim P(x) = N(0, 1)$$

$$y \sim P(y|x)$$

$$P(y = 1|x) = \sigma(3x - 1) = \frac{1}{1 + e^{-3x+1}} \in (0, 1)$$

$$P(y = 0|x) = 1 - P(y = 1|x)$$

Examples for such abstract $(x, y) \sim P(x, y)$ in binary classification:

- Input space $X = \mathbb{R}^2$, Output Space $\{-1, +1\}$

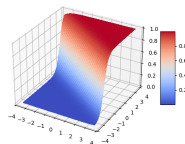
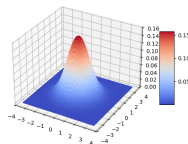
$$x = (x^{(1)}, x^{(2)}) \sim N\left(\mu = (0, 0), \Sigma = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right)$$

$$y \sim P(y|x)$$

$$P(y = 1|x) = \sigma(2x^{(1)} + 3x^{(2)} - 1) \in (0, 1)$$

$$P(y = 0|x) = 1 - P(y = 1|x)$$

- defines $P(x)$ and $P(y = 1|X = x)$.
- allows to draw samples step by step. How to get $P(x, y)$ from that?



$$x = (x^{(1)}, x^{(2)}) \sim N\left(\mu = (0, 0), \Sigma = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right)$$
$$y \sim P(y|x), \quad P(y = 1|x) = \sigma(2x^{(1)} + 3x^{(2)} - 1) \in (0, 1)$$

How to get $P(x, y)$ from that?

- by the definition of conditional probabilities:

$$P(x, y) = P(y|x)p(x) \quad \text{done!}$$

- insight: when we draw $x \sim P(x)$, $y \sim P(Y|X = x)$, then $P(y, x)$ is fully specified already

- useful: gives an idea how to draw samples from $P(x, y)$ when one specifies $(P(x), P(y|x))$ or $(P(y), P(x|y))$
- have $P(x, y) = P(x)P(y|x)$ from known $P(x), p(y|x)$
 - draw $x \sim P(x)$
 - draw $y \sim p(y|X = x)$ where you condition on your drawn x
 - obtain sample (x, y) drawn from $p(x, y)$
- use this to generate synthetic data to test algorithms

- useful: gives an idea how to draw samples from $P(x, y)$ when one specifies $(P(x), P(y|x))$ or $(P(y), P(x|y))$
- **reverse way:** also have $P(x, y) = P(y)P(x|y)$ from known $P(y), P(x|y)$
 - draw $y \sim P(y)$
 - draw $x \sim p(x|Y = y)$ where you condition on your drawn y
 - obtain sample (x, y) drawn from $p(x, y)$
- use this to generate synthetic data to test algorithms

- ① Probability distributions
- ② Modelling future unforeseeable data points
- ③ The goal of training machine learning / AI systems
- ④ Why one MUST NEVER use test data to select/optimize your ML system
- ⑤ Overfitting

We had before:

the goal of machine learning described in words

To find a prediction model, which predicts well on unseen and unforeseeable (unvorher-sagbar) data points

- We want a model which predicts well on $(x, y) \sim P$.

What does it mean to **predict well**?

- low value of the loss function L on $(x, y) \sim P$!
- can be expressed as expected loss under data drawn from P :

$$E_{(x,y) \sim P}[L(f(x), y)]$$

- The loss function $L(f(x), y)$ measures the deviation between prediction $f(x)$ and ground truth y , for a given sample (x, y)

the goal of machine learning specified further

For a given loss function L , to find a prediction model f such that under the probability distribution for samples with their labels (x, y) the expected loss

$$E_{(x,y) \sim P}[L(f(x), y)]$$

is low

Low expected loss means: **errors are not prevented**. Predictions will make mistakes.

For binary classification one would choose for example the 0-1 loss

$$L(f(x), y) = \mathbb{1}[f(x) \neq y] = \begin{cases} 1 & f(x) \neq y \\ 0 & f(x) = y \end{cases}$$

$$E_{(x,y) \sim P(x,y)}[\mathbb{1}[f(x) \neq y]] = 1 P(f(x) \neq y) + 0 P(f(x) = y) = P(f(x) \neq y)$$

- in real ML problems: you do not know the data-generating $P(x, y)$
- challenge of ML: want

$$E_{(x,y) \sim P}[L(f(x), y)]$$

to be low, but do not know P (+ cannot compute this loss).

important question

How to know whether the expected loss for a given f is high or low

... when we cannot compute it exactly???

- assume that we have drawn some samples $T = ((x_0, y_0), \dots, (x_{n-1}, y_{n-1}))$ such that $(x_i, y_i) \sim P$
- we could compute the mean of the loss on them:

$$\hat{E}(L, f, T) = \frac{1}{n} \sum_i L(f(x_i), y_i)$$

- in what way does $\hat{E}(L, f, T)$ inform us about $E_{(x,y) \sim P}[L(f(x), y)]$?

the central limit theorem

Let $Z_0, Z_1, \dots, Z_{n-1}, \dots$ be an (infinite) sequence of iid (identically distributed and independent) random variables. Let $E[Z_i] = m$ and $\text{Var}[Z_i] = \sigma^2 < \infty$. Consider $\hat{e} = \frac{1}{n}(\sum_i^n Z_i)$ to be the empirical mean (Mittelwert)

Then the random variable $\sqrt{n}(\hat{e} - m)$ converges in distribution to a normal distribution $N(0, \sigma^2)$

(Convergence in distribution is convergence of cumulative distribution functions.)

- we apply it to: $Z_i = L(f(X_i), Y_i)$ – the loss of the predictor on a sample
- $E[Z_i] = m = E[L(f(X_i), Y_i)]$ – what we want to know.
- need to clarify what $\sqrt{n}(\hat{e} - m)$ tells us about $\frac{1}{n}(\sum_i^n Z_i)$!?

One thing to understand: $\sqrt{n}(\hat{e} - m)$ converges in distribution to a normal distribution $N(0, \sigma^2)$

- Assume that the random variable A with $A = \sqrt{n}(\frac{1}{n}(\sum_i^n Z_i) - m)$ would be exactly normally distributed as $N(0, \sigma^2)$, then what would be the distribution of $\frac{1}{n}(\sum_i^n Z_i)$?

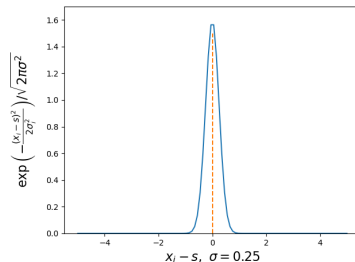
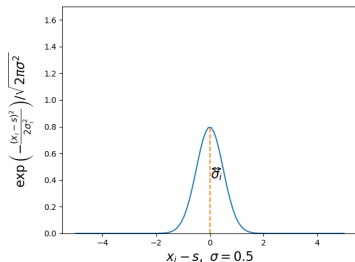
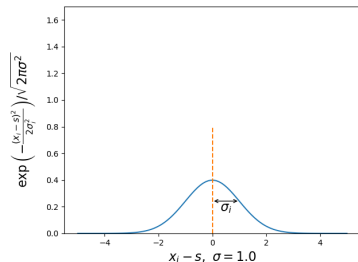
$$\begin{aligned} A &= \sqrt{n}(\hat{e} - m) \\ \Rightarrow \frac{A}{\sqrt{n}} &= \hat{e} - m \\ \Rightarrow \frac{A}{\sqrt{n}} + m &= \hat{e} = \frac{1}{n}(\sum_i^n Z_i) \end{aligned}$$

- we are interested to know how $\hat{e} = \frac{1}{n}(\sum_i^n Z_i)$ is distributed
- it is a an affine transformation of a random variable $A \sim N(0, \sigma^2)$

if $A \sim N(0, \sigma^2)$ and $B = cA + m$, then B is also normally distributed with the following parameters: $B \sim N(m, c^2\sigma^2)$

- $\hat{e} = \frac{1}{n}(\sum_i^n Z_i)$ is normally distributed with mean $m = E[L(f(X), Y)]$ and variance σ^2/n , that is $\frac{1}{n}(\sum_i^n Z_i) \sim N(m, \sigma^2/n)$
- $m = E[L(f(X), Y)]$ the uncomputable quantity we want to know about
- what is the impact of a variance σ^2/n ?

Lets look at normal distributions with $\sigma^2/n \rightarrow 0$ as n increases:



- as $\sigma^2/n \rightarrow 0$ the distribution converges to one point, namely $m = E[L(f(X), Y)]$
- as we get larger test sets, the probability that our estimate $\hat{e} = \frac{1}{n}(\sum_i^n L(f(x_i), y_i))$ is close to the uncomputable $E[L(f(X), Y)]$, increases towards 1.
- as we get larger test sets, the probability that our estimate $\hat{e} = \frac{1}{n}(\sum_i^n L(f(x_i), y_i))$ has a large deviation from the uncomputable $E[L(f(X), Y)]$, decreases towards 0.

- use data $\{(x^{\{1\}}, y^{\{1\}}), \dots, (x^{\{n\}}, y^{\{n\}})\}$ which one never have used to select the function f
- if $(x^{\{i\}}, y^{\{i\}}) \sim P$ and drawn statistically independently, and if the variance of the loss $\sigma^2 = E[L(f(x), y)^2] - (E[L(f(x), y)])^2 < \infty$,

then by the **central limit theorem**, we have that

$$\hat{E}(L, f, T) = \frac{1}{n} \sum_{i=1}^n L(f(x^{\{i\}}), y^{\{i\}})$$

is **approximately (!!)** normally distributed with mean $E_{(x,y) \sim P}[L(f(x), y)]$ and variance $\frac{\sigma^2}{n}$, where σ^2 is the variance of L under the unknown P .

- $E_{(x,y) \sim P}[L(f(x), y)]$ is what we want to know!

Under some conditions, by the **central limit theorem**, we have that

$$\hat{E}(L, f, T) = \frac{1}{n} \sum_{i=1}^n L(f(x^{\{i\}}), y^{\{i\}})$$

is **approximately (!!)** normally distributed with mean $E_{(x,y) \sim P}[L(f(x), y)]$ and variance $\frac{\sigma^2}{n}$, where σ^2 is the variance of L under the unknown P .

Consequence ?

- $\frac{\sigma^2}{n} \rightarrow 0$ as $n \rightarrow \infty$. This means, the normal distribution collapses onto a point centered at the mean $E_{(x,y) \sim P}[L(f(x), y)]$
- this means that for a sufficiently large test set,

$$\hat{E}(L, f, T) = \frac{1}{n} \sum_{i=1}^n L(f(x^{\{i\}}), y^{\{i\}})$$

will be close to the uncomputable $E_{(x,y) \sim P}[L(f(x), y)]$

However, we **cannot say exactly** how close for a given n !

There are so-called concentration inequalities which give bounds for **large deviations** of

$$|\hat{E}(L, f, T) - E_{(x,y) \sim P}[L(f(x), y)]|$$

which are valid with a certain probability to be true, e.g.

https://en.wikipedia.org/wiki/Hoeffding%27s_inequality,

however these bounds are too conservative for practical machine learning tasks

The normal distribution has infinite tails

- ⊙ !! there is non-zero probability, that the test set T might result in an average loss $\hat{E}(L, f, T)$ which is far off the expected loss

$$\hat{E}(L, f, T) \gg \text{or} \ll E_{(x,y) \sim P}[L(f(x), y)]$$

- ⊙ $\frac{\sigma^2}{n} \rightarrow 0$ as $n \rightarrow \infty$... as the test set gets larger, the probability of such large deviations goes to zero.
- ⊙ we can approximately estimate the probability that the expected loss is far off the average loss $\hat{E}(L, f, T)$ which we have computed on n test samples - by using the cumulative density function https://en.wikipedia.org/wiki/Cumulative_distribution_function of the normal distribution

Measuring loss on test data not used in training

If you use only data, which was never used for training and hyperparameter selection, and for a sufficiently large sample size n , then the loss

$$\hat{E}(L, f, D_n) = \frac{1}{n} \sum_{i=1}^n L(f(x^{\{i\}}), y^{\{i\}})$$

is an approximation for the performance on new, unseen data points.

Therefore, if the average loss on test data, never used in training/hyperparameter selection is small, then we can be satisfied with the prediction function f

though there is a small probability that there is a large deviation between the average loss $\hat{E}(L, f, T)$ and the expectation of the loss under the probability which generates the data

Example:

$$E[\mathbb{1}(f(X) \neq Y)] = P(f(X) \neq Y) \approx \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{1}(f(x_i) \neq y_i)$$

is an approximation which would get better if we collect more test data , provided that the sample collection is done statistically independently

(remember: knowing the value of one variable does not provide information about the distribution of another variable)

The key challenge in machine learning

- We want to be able to **predict well on unseen and unforeseeable data** - the data which we expect when the model is deployed.
- In real ML problems we do not know from what distribution that future data to be seen in deployment is coming.
- Instead we have a finite set of data samples $\{(x^{\{1\}}, y^{\{1\}}), \dots, (x^{\{n\}}, y^{\{n\}})\}$ from the unknown distribution to ensure that we do well on unseen and unforeseeable data.
- goal is to find a mapping f such that its (uncomputable) expected loss is close to the best possible one: $E[L(f(x), y)] \approx \inf_{g \in \mathcal{F}} E[L(g(x), y)]$

How to do machine learning in practice ?

- collect a set of data points $S = (x_0, y_0), \dots, (x_{n-1}, y_{n-1})$, such that they are approximately statistically independent
- split the dataset S into **disjoint!!** training, validation and test sets –
- train on the training set
- select hyperparameters on the validation set
- at the end, measure the performance on the test set

- ① Probability distributions
- ② Modelling future unforeseeable data points
- ③ The goal of training machine learning / AI systems
- ④ Why one MUST NEVER use test data to select/optimize your ML system
- ⑤ Overfitting

The DONOT of ML

Never use test data to train your model or to choose hyperparameters

- If your model development sees and uses the test data, then you do not measure the loss on unseen and unforeseeable data anymore.
- You have no guarantee anymore to **predict well on unseen and unforeseeable data**
- changes to: to predict well on data which you have seen before

Why we cannot check the performance on test data to improve our selection of mappings?

| 56

Why never use test data for training and hyperparameter selection?

The simple answer:

Why it is forbidden to check on test data

- ⊙ We want a guarantee that our predictions are good on unseen and unforeseeable data samples.

If we use data used in training or hyperparameter selection, then we have a guarantee for partially seen data samples - this is not a useful guarantee for models to be deployed.

The precise reason why to use only data, which was never used for training and hyperparameter selection ... is subtle: **Central limit theorem does not hold anymore** / its conditions are not met anymore. For those who want to know, it is explained out of class at the end of these slides.

Why we cannot check the performance on test data to improve our selection of mappings?

| 57

Why never use test data for training and hyperparameter selection?

If f is a constant function, not depending on the data, then the random variable $Z_i = L(f(X_i), Y_i)$ is drawn from the same distribution for all test datasets T_n . Then we can apply the central limit theorem for the limit $n \rightarrow \infty$.

However, if f is chosen using data sets which contains test samples (X_i, Y_i) from T_n , then f becomes a function of the test data. In that case the distribution of

$$L(f(X_i), Y_i) = L(f[T_i](X_i), Y_i)$$

changes for every n , because

- we use T_n, T_{n+1}, T_{n+2} etc to select f
- when we take the limit $n \rightarrow \infty$ in the CLT, we might choose a different f for some T_{n+k} than for T_n (even if it happens only for some draws of test sets)

Consequence: $Z_i = L(f[T_i](X_i), Y_i)$ is not identically distributed anymore. **Central limit theorem does not apply in this case anymore**

Why we cannot check the performance on test data to improve our selection of mappings?

| 58

One can show even a stronger result:

Let $\mathcal{F}$ be a set of mappings from which we select one by the training algorithm. Then the following always holds

$$E[\inf_{f \in \mathcal{F}} L(f(X), Y)] \leq \inf_{f \in \mathcal{F}} E[L(f(X), Y)]$$

What does that mean?

- $\inf_{f \in \mathcal{F}} L(f(X), Y)$ selects the best mapping for a given random draw (x, y) – in this case f changes with X, Y . Then we compute the average $E[\cdot]$ over all (x, y)
- $\inf_{f \in \mathcal{F}} E[L(f(X), Y)]$ – this is the best **single unchanging** mapping which we can choose from $\mathcal{F}$ if we had a perfect training algorithm
- When we select the best mapping for each sample (x, y) , then the expectation over this is *ALWAYS* optimistic:

$$\underbrace{E[\inf_{f \in \mathcal{F}} L(f(X), Y)]}_{\text{maximal cheating}} \leq \underbrace{\inf_{f \in \mathcal{F}} E[L(f(X), Y)]}_{\text{best possible constant choice}}$$

and therefore an uninformative estimate

Why we cannot check the performance on test data to improve our selection of mappings?

| 59

Why we cannot check the performance on test data to improve our selection of mappings

Why only use only data, which was never used for training and hyperparameter selection?

the short version: we lose our performance guarantees on new unseen data points (x, y)

[https://www.creativebloq.com/ai/ai-art/
forget-ai-image-generators-an-autonomous-ai-artist-just-sold-usd351-600-in-paintings-at-sothebys](https://www.creativebloq.com/ai/ai-art/forget-ai-image-generators-an-autonomous-ai-artist-just-sold-usd351-600-in-paintings-at-sothebys)

- ① Probability distributions
- ② Modelling future unforeseeable data points
- ③ The goal of training machine learning / AI systems
- ④ Why one MUST NEVER use test data to select/optimize your ML system
- ⑤ Overfitting

setup:

- we have some training data set D of samples (x, y) and possibly a test set T of different samples.
- we want to use the training data set to train a model, that is to select a mapping $f : \mathcal{X} \rightarrow \mathcal{Y}$ from a set $\mathcal{F}$ of all possible mappings.

Next: try to understand by examples what overfitting is

informally: one chooses a mapping by training on data which is not the best in the sense of expected loss

The key challenge in machine learning

- We want to be able to **predict well on unseen and unforeseeable data** - the data which we expect when the model is deployed.
- In real ML problems we do not know from what distribution P the future data in deployment will be coming from.
- Instead we have a finite set of data samples $D = \{(x^{\{1\}}, y^{\{1\}}), \dots, (x^{\{n\}}, y^{\{n\}})\}$ from the unknown distribution to ensure that we do well on unseen and unforeseeable data.
- goal is to find a mapping f such that its (uncomputable) expected loss is close to the best possible one: $E[L(f(x), y)] \approx \inf_{g \in \mathcal{F}} E[L(g(x), y)]$

informally: one chooses a mapping by training on data which is not the best in the sense of expected loss

$$f^{(ov)} = \text{Training}(D)$$

$$E[L(f^{(ov)}(x), y)] > \inf_g E[L(g(x), y)]$$

- challenge of ML: want

$$E_{(x,y) \sim P}[L(f(x), y)]$$

to be low, but do not know P (+ cannot compute this expectation).

- Have instead a few data samples $\{(x^{\{1\}}, y^{\{1\}}), \dots, (x^{\{n\}}, y^{\{n\}})\}$ drawn from P

Next step:

- show that even for very simple ML problems ... one has a substantial probability to choose a model f with high error under P – when one makes the choice based on a finite set of samples.

The setup of our next example:

- we can observe outcomes of an unfair coin, heads or tails
- goal: predict which the most likely outcome: 0 or 1?
- binary classification problem: 2 outcome states $Y = \{0, 1\}$
- have no features here to use, but we have some data samples $y^{\{1\}}, \dots, y^{\{n\}}$ drawn from the unknown distribution P
 - we assume that the mechanism deciding the outcome stays the same over time, no hidden changes over time
- goal: estimate $P(Y = 0) > 0.5$ or $P(Y = 0) < 0.5$ from observations of the coin $y^{\{1\}}, \dots, y^{\{n\}}$
- we can predict only two models here: $f_0(x) = 0$ or $f_1(x) = 1$

- we can predict only two models here: $f_0(x) = 0$ or $f_1(x) = 1$
- lets assume we would know what P is: $P(y = 0) = 0.4$. Then we do know which one is the best model.
- what is the best model to predict – in the sense of lowest expected loss $E[L(f(x), y)]$?
- use classification loss $L(f(x), y) = 1[f(x) \neq y]$. Then:

$$\begin{aligned} E[L(f_0(x), y)] &= 1[f_0(x) \neq y](0)p(y = 0) + 1[f_0(x) \neq y](1)p(y = 1) \\ &= 1[0 \neq y](1)p(y = 1) = 0.6 \end{aligned}$$

$$E[L(f_1(x), y)] = 0.4$$

- Therefore the optimal model to be chosen is $f_1(x)$ (in the sense of lowest expected loss)

- $P(y = 0) = 0.4$. The optimal prediction is $f_1(x)$
- if we draw one sample $y \sim P(y)$, and if we would base our decision to select f_0 or f_1 on the data (y) we have seen, what is the probability to choose the wrong model f_0 ?
- this happens if we observe $y = 0$. $P(y = 0) = 0.4$ This occurs 40% of the time!!

Take away:

even in simple problems we have a high chance to decide for the suboptimal model (which has not the lowest loss) if we decide based on a finite set of data samples

Intuition: the randomness when collecting data can lead to data sets which are misleading about the true distribution of samples $P(x, y)$...

a finite set of random samples can be non-representative about $P(x, y)$

We still assume $P(y = 0) = 0.4$. The optimal prediction in the sense of lowest expected loss is $f_1(x)$

- if we draw three samples $y_1, y_2, y_3 \sim P(y)$, and if we would base our decision on the data we have seen, what is the probability to choose the wrong model f_0 ?
- one can check the sum of the three samples and make a majority decision based on $\sum_{i=1}^3 y_i$
- we make the wrong decision if two out of the three will be $y_i = 0$. That is if we observe 100, 010, 001, 000
- what is the probability for that ?

- if we draw three samples $y_1, y_2, y_3 \sim P(y)$, and if we would base our decision on the data we have seen, what is the probability to choose the wrong model f_0 ?
- we make the wrong decision if two out of the three will be $y_i = 0$. That is if we observe 100, 010, 001, 000
- what is the probability for that ?
 - $P(100) = (1 - 0.4) * 0.4 * 0.4 = 0.4^2(1 - 0.4)^1$, $P(100, 010, 001) = \binom{3}{1} 0.4^2(1 - 0.4)^1$
 - repeated experiments, binary outcome in each, independent, same probability for outcome
 - Binomial distribution with $n = 3, p = 0.4$ and outcomes of at least 2.

$$P_{n,p}(k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

$$r = \sum_{k=2}^3 P_{n,p}(k)$$

- Whenever we observe one of these sets of three samples: 100, 010, 001, 000, we will choose the suboptimal prediction model $f_0(x)$.

One can show:

$$P(\{100, 010, 001, 000\}) = \sum_{k=2}^3 P_{3,0.4}(k) \approx 0.35 < 0.4$$

- still a large probability to choose the poorly performing prediction $f_0(x)$.

- Same setup, but draw 5 samples. Probability to choose the non-optimal model, if we observe at least 3 of the 5 to be $y_i = 0$.

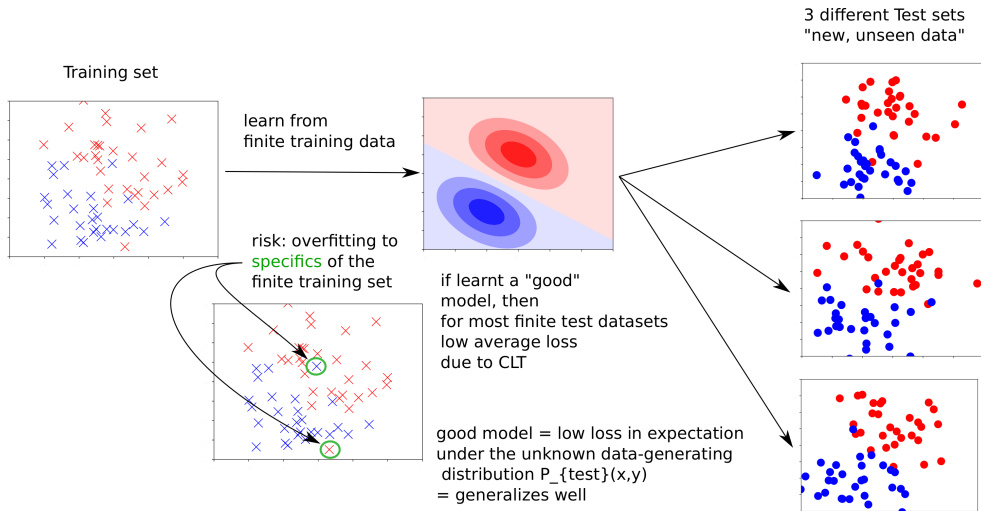
One can show:

$$P\left(\sum_{i=1}^5 y_i \leq 2\right) = \sum_{k=3}^5 P_{5,0.4}(k) \approx 0.317 < 0.35$$

- still a large probability to choose the poorly performing prediction $f_0(x)$.
- probability to choose the non-optimal model decreases as the training sample size increases.

- We had this problem : deciding based on some observations $y^{\{1\}}, \dots, y^{\{n\}}$ whether the most likely outcome is $y = 0$ or $y = 1$ (and thus whether we should choose f_0 or f_1).
- Observation: Even in the simplest problems , if you have some randomness in your data, then you could end up choosing a suboptimal model with a not so small probability.

- When randomness is present, then the finite set of samples $y^{\{1\}}, \dots, y^{\{n\}}$ can be “not fully representative” for your distribution P (in our example, whenever we observe more $y^{\{i\}} = 0$ than $y^{\{i\}} = 1$ despite $P(Y = 0) = 0.4$)



How is above problem related to classification problems?

- in the above problem we are estimating $P(Y = 0)$ in a single feature point x
- binary classification problems: we want to estimate $P(Y = 0|X = x)$ for all possible features x , not only $P(Y = 0)$ in a single point.

generalizing the setup:

- We have a class of prediction models $\mathcal{F} = \{f_0, f_1, f_2, f_3, \dots\}$
- we have a loss function $L(f(x), y)$
- We want to choose $f_i \in \mathcal{F}$ with minimal expected loss under the distribution P

$$\min_{f \in \mathcal{F}} E_{(x,y) \sim P}[L(f(x), y)]$$

(we assume for simplicity that $\mathcal{F}$ is countable)

Consider the setup where we have a loss L , some data, and a set of prediction functions $f \in \mathcal{F}$ to choose from. $\mathcal{F}$ contains only finitely many functions.

The best possible prediction is to choose an f^* such that its expected loss is minimal:

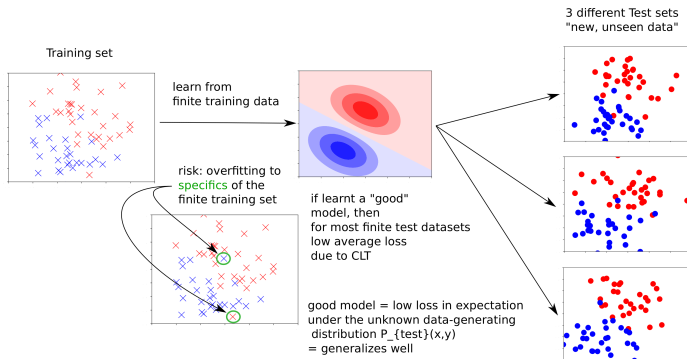
$$f^* = \operatorname{argmin}_{g \in \mathcal{F}} E_{(x,y) \sim P}[L(g(x), y)]$$

Overfitting (finite function class)

Overfitting occurs, if we choose a function $f^{(ov)}$ which has a higher expected loss than the best possible function:

$$E_{(x,y) \sim P}[L(f^{(ov)}(x), y)] > \min_{g \in \mathcal{F}} E_{(x,y) \sim P}[L(g(x), y)]$$

This contains two aspects, where finite randomly drawn samples (x, y) can be misleading:



- training:
 - training data - could lead to too much overfitting
 - validation data - selection of the wrong hyperparameters
- testing data - could lead to an estimate of the expected loss (via the average on test samples) which is far off the true expectation of the loss